Project Phase 1: The Launch and Global Presence

Due Date: End of Week 3 **Points:** 100

Scenario

Your team has been hired as cloud engineers/architects for a small local business, "The Riverside Cafe", owned by David and Helen. The Riverside Cafe has a great local reputation but currently has no online presence. Their daughter, Emily, wants to create a simple website to showcase their desserts and coffees, and provide essential business information like location, hours, and contact details.

A web developer has designed a static website for Emily and will send you the files. However, Emily has a few concerns: Since David and Helen aren't very technical, Emily is worried about accidentally losing the website's content. As such, she wants to keep a history of all website versions and updated content. To avoid increasing storage costs, she wants to keep old versions of the files for only one month. In addition, the website is critical for the business, so it must be restorable if the primary cloud region experiences an outage.

Your first task is to establish this initial web presence. The goal is to launch a simple, reliable, and cost-effective static website while establishing best practices from the very beginning.

Project Goals

1. **Launch a Public Website.** The Riverside Cafe needs a publicly accessible website. The initial content will be static (HTML, CSS, images).
2. **Protect Website Data.** The website content will be updated frequently. A strategy must be in place to prevent accidental deletion or corruption of the website's files.
3. **Plan for the Future.** The solution must consider long-term cost management for storing multiple versions of files and have a basic disaster recovery plan.

Technical Requirements

- You are free to use any major cloud provider (AWS, Azure, Google Cloud).
- All resources you create must be documented.
- The provided source code for the website (`index.html` and an `images` folder) will be given to you. You are free to modify or enhance it.

Submission Instructions

You will submit a single PDF document to Moodle. This document must contain all parts of the deliverable outlined below. Please use clear headings for each section.

Deliverable Breakdown

- **Team Charter:** Include your completed team charter based on the Teamwork Guideline document provided in class.
- Provide the public, clickable URL to your live and functioning static website.
- Provide your Cloud Architecture Design.
- Provide a Data Protection Plan.
- Provide a Cost Optimization Plan.
- Provide a Disaster Recovery (DR) Plan.
- Provide a Justification. State which cloud provider and which specific object storage service (e.g., Amazon S3, Azure Blob Storage) you chose. Explain why this service is the ideal choice for hosting a static website compared to other options, such as using a virtual machine. Explain the steps you took to make the website's content publicly readable while keeping the storage administration secure. Include a code snippet of any access policies you created (e.g., an S3 bucket policy) and explain what the key statements in the policy do. Provide a high-level, step-by-step description of how you deployed the website files to your chosen storage service.

Grading Rubric

Criteria	Description	Points
Team Charter	The team charter is submitted and adheres to the provided guidelines.	10
Live Website	The provided URL leads to a functioning static website displaying The Riverside Cafe's content.	10
Architecture Justification	Clear justification for the choice of cloud provider and service. Explanation of security configuration is logical and demonstrates an understanding of public access vs. private administration.	20
Data Protection Plan	A suitable data protection feature (e.g., versioning) is identified, implemented, and clearly explained as a solution to the business problem.	15
Cost Optimization Plan	A logical lifecycle policy is designed and justified with respect to the business context.	20

Disaster Recovery Plan	A suitable cross-region replication strategy is implemented and the failover concept is explained.	15
Professionalism & Clarity	The submitted document is well-organized and clear. It also clearly discusses the team's working communication and individual contributions for the phase.	10
Total		**100**